

Lead / lag problems

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

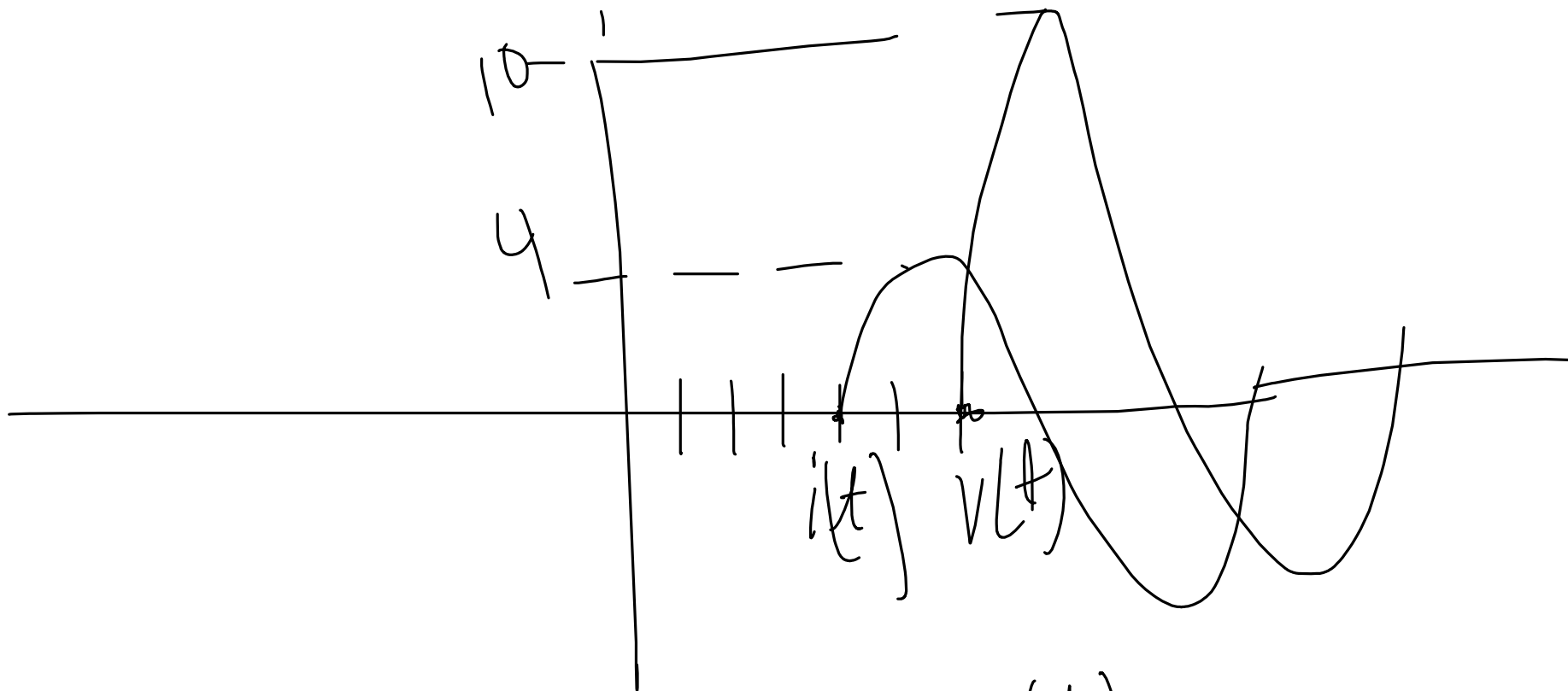
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

(a) $v(t) = 10 \cos(4t - 60^\circ)$ and
 $i(t) = 4 \sin(4t + 50^\circ)$

$$\begin{aligned} i(t) &= 4 \sin(4t + 50^\circ) \\ &= 4 \cos(4t + 50^\circ - 90^\circ) \\ &= 4 \cos(4t - 40^\circ) \end{aligned}$$



$\therefore i(t)$ leads $v(t)$

$$v_2(t) = 20 \cos 5t \text{ V}$$

(c) $x(t) = 13 \cos 2t + 5 \sin 2t$ and
 $y(t) = 15 \cos(2t - 11.8^\circ)$

$$5 \sin(2t) = 5 \cos(2t - 90^\circ)$$

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

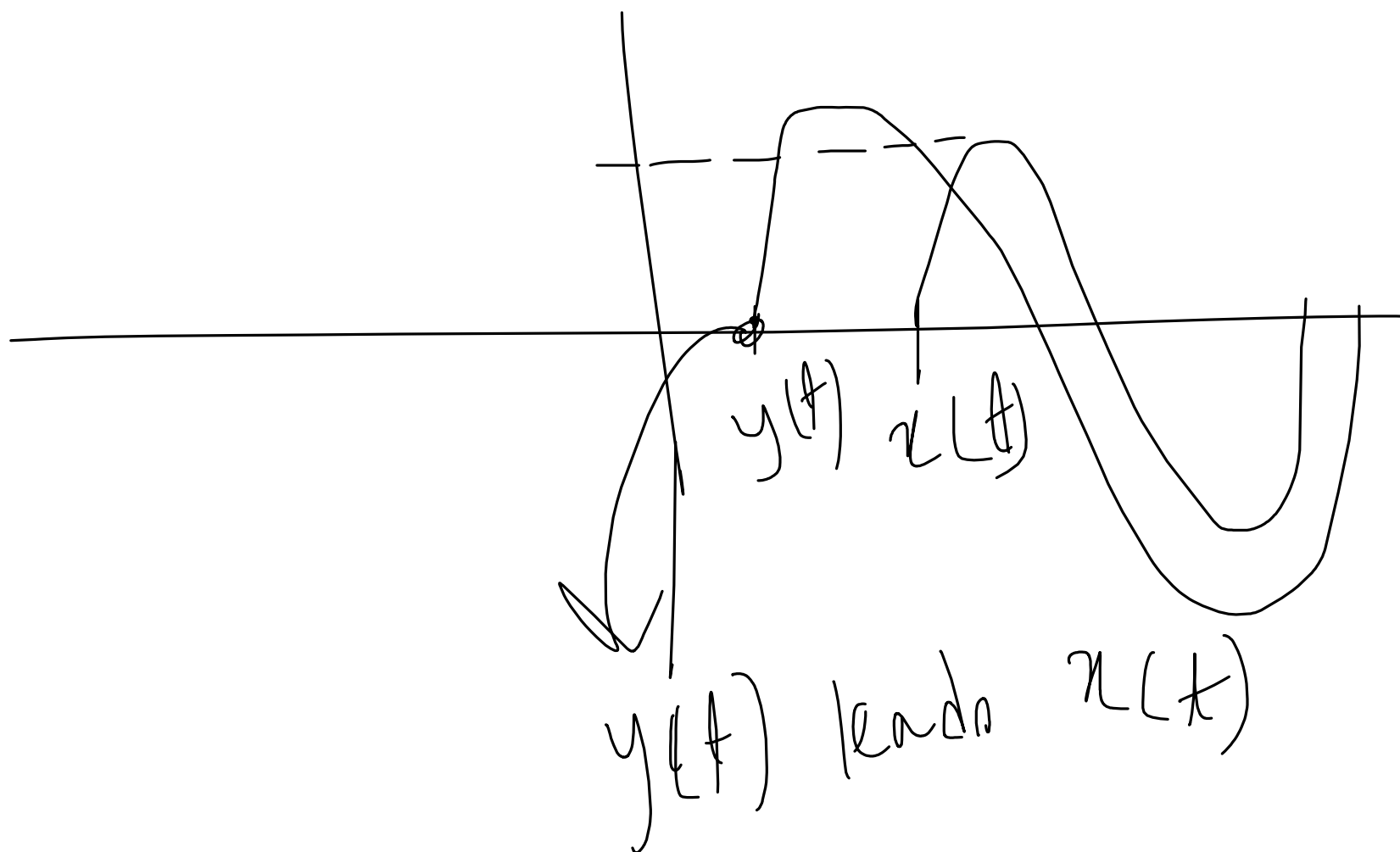
$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$X = 13 \angle 0^\circ + 5 \angle -90^\circ$$

$$= 13.93 \angle -21.04^\circ$$

$$x(t) = 13.93 \cos(2t - 21.04^\circ)$$



9.18 Obtain the sinusoids corresponding to each of the following phasors:

(a) $V_1 = 60 \angle 15^\circ \text{ V}, \omega = 1$

(b) $V_2 = 6 + j8 \text{ V}, \omega = 40$

(c) $I_1 = 2.8e^{-j\pi/3} \text{ A}, \omega = 377$

(d) $I_2 = -0.5 - j1.2 \text{ A}, \omega = 10^3$

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi)$$

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$

(c)

$$I_1 = 2.8 \angle -j\pi/3$$

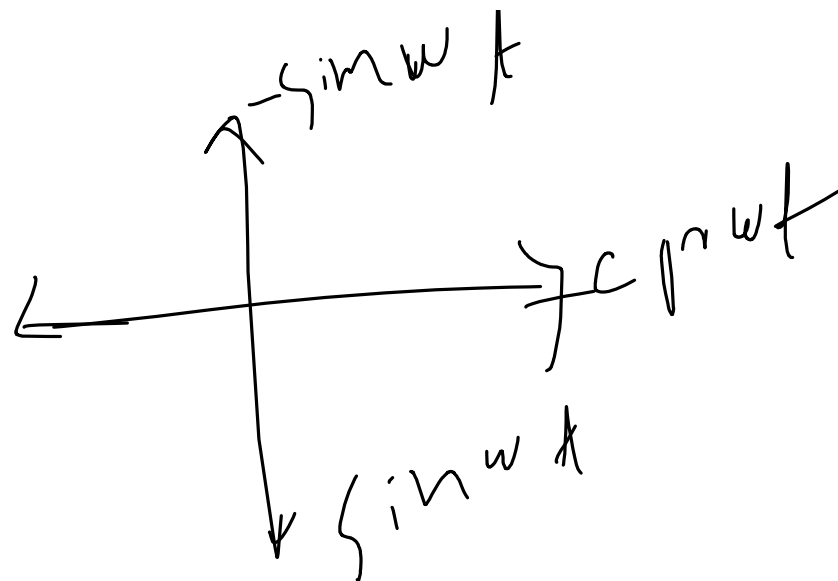
$$= 2.8 \times (\cos(\pi/3) - j \sin(\pi/3))$$

$$= 2.8 \times \left(\frac{1}{2} - j \frac{\sqrt{3}}{2} \right)$$

$$= \underline{\underline{1.4 - j 2.4^3}}$$

$$= 2.8 \angle -60.05^\circ$$

$$i_1(t) = 2.8 \cos(377t - 60.05^\circ)$$



9.19 Using phasors, find:

(a) $3 \cos(20t + 10^\circ) - 5 \cos(20t - 30^\circ)$

(b) $40 \sin 50t + 30 \cos(50t - 45^\circ)$

(c) $20 \sin 400t + 10 \cos(400t + 60^\circ) - 5 \sin(400t - 20^\circ)$

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

Handwritten solution for (c):

$$x_1 = 20 \cos(400t - 90^\circ)$$

$$x_2 = 5 \sin(400t - 20^\circ)$$

$$= 5 \cos(400t + 90^\circ - 20^\circ)$$

$$= 5 \cos(400t + 70^\circ)$$

$$\therefore \cancel{x} = x_1 + x_2 + x_3$$

$$= 20 \angle -90^\circ + 5 \angle 70^\circ + 10 \angle 60^\circ$$

$$\rightarrow = 9.44 \angle -44.7^\circ$$

$$\therefore x(t) = 9.44 \cos(400t - 44.7^\circ)$$

9.21 Simplify the following:

(a) $f(t) = 5 \cos(2t + 15^\circ) - 4 \sin(2t - 30^\circ)$

(b) $g(t) = 8 \sin t + 4 \cos(t + 50^\circ)$

(c) $h(t) = \int_0^t (10 \cos 40t + 50 \sin 40t) dt$ ✓

$50 \sin(40t - 90^\circ)$

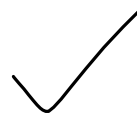
$$\int x dt = \frac{x}{j\omega}$$

$$H(A) = \frac{1}{j\omega} (10 \angle 0^\circ + 50 \angle -90^\circ)$$
$$\frac{10 \angle 0^\circ + 50 \angle -90^\circ}{j40}$$

$$= 1.275 \angle -168.7^\circ \checkmark$$



$$1.275 \angle \cos(40^\circ - 168.7^\circ)$$



✓
9.24 Find $v(t)$ in the following integrodifferential equations using the phasor approach:

(a) $v(t) + \int v \, dt = 10 \cos t$

(b) $\frac{dv}{dt} + 5v(t) + 4 \int v \, dt = 20 \sin(4t + 10^\circ)$

$\frac{dx}{dt} \rightarrow j\omega x$
 $\int x \, dt \rightarrow \frac{x}{j\omega}$
 $20 \cos(4t + 10 - 90^\circ)$

$$j\omega V + 5V + \frac{4V}{j\omega} = 20 \angle -80^\circ$$

$\omega = 4$

(b)

$$V \left(j\omega + 5 + \frac{4}{j\omega} \right) = 20 \angle -90^\circ$$

$$\Rightarrow V \left(j4 + 5 + \frac{4}{j4} \right) = 20 \angle -90^\circ$$

$$\Rightarrow V = \frac{20 \angle -90^\circ}{(j4 + 5 + \frac{1}{j})}$$

$$V = 3.42 \angle -110.7^\circ$$

$$v(t) = 3.42 \cos(4t - 110.7^\circ)$$

through the resistor and the capacitor.

9.31 A series RLC circuit has $R = 80 \Omega$, $L = 240 \text{ mH}$, and $C = 5 \text{ mF}$. If the input voltage is $v(t) = 10 \cos 2t$, find the current flowing through the circuit.

$V = 10 \angle 0^\circ$

$$X_L = j\omega L = j2 \times 240 \text{ m} = j0.48$$

$\frac{1}{j}$

$$X_C = \frac{1}{j\omega C} = -j \times \frac{1}{2 \times 5 \text{ m}} = -j100$$

$$Z_t = R + X_L + X_C$$

$$= 80 + j0.48 - j100$$

$$= 80 - j99.52$$

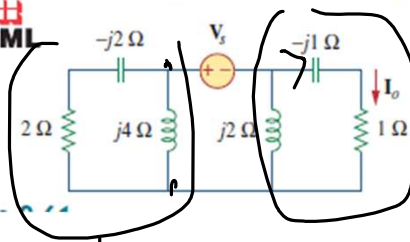
$$I = \frac{V}{Z_t}$$

$$= 0.078 \angle 51.21^\circ$$

$$i(t) = 0.078 \cos(2t + 51.21^\circ)$$

CKA

9.54 In the circuit of Fig. 9.61, find V_s if $I_o = 2\angle 0^\circ$ A.



z_1

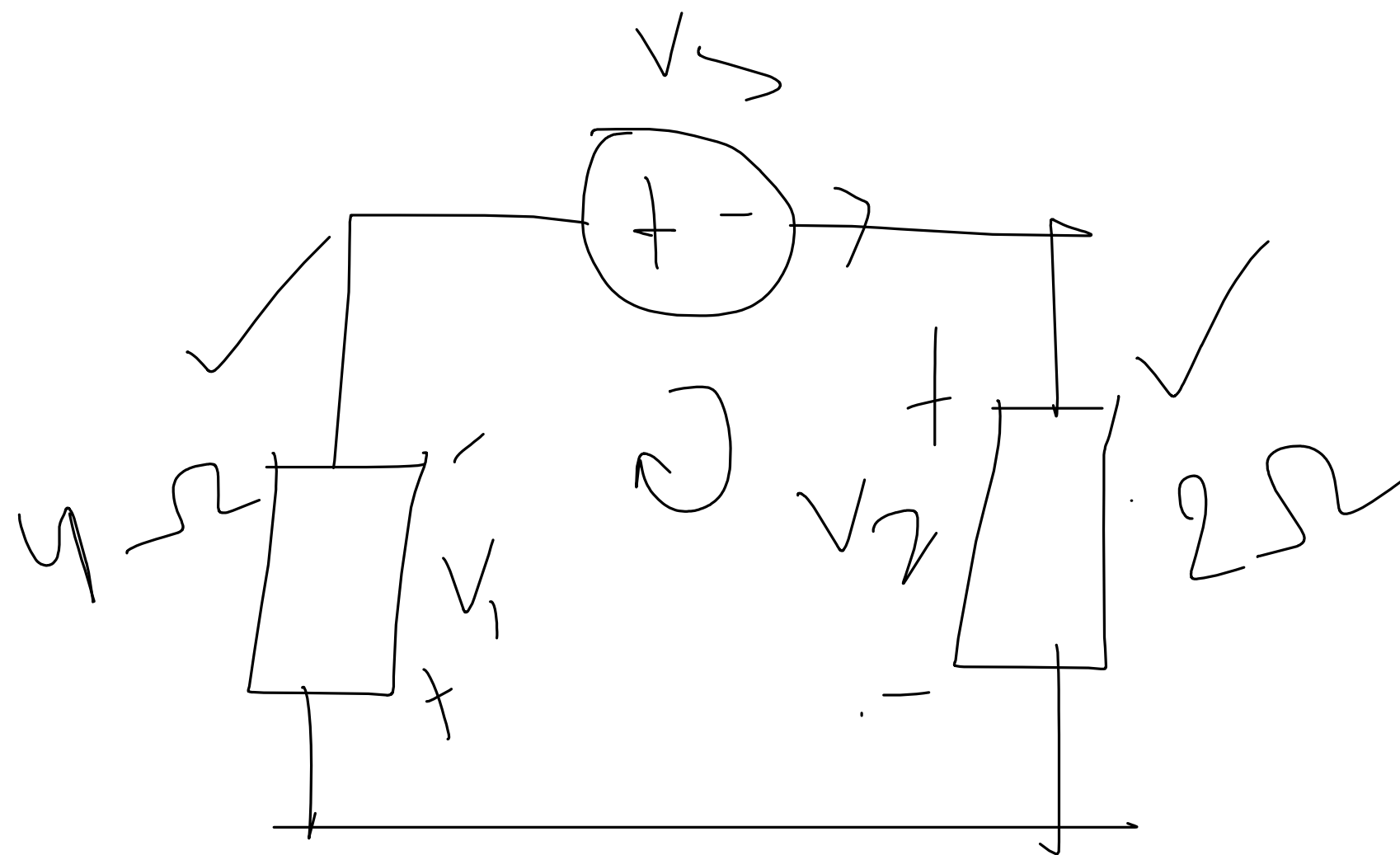
$$z_1 = (2 - j2) \parallel j4$$

$$= \frac{(2 - j2) \times j4}{2 - j2 + j4} = 4$$

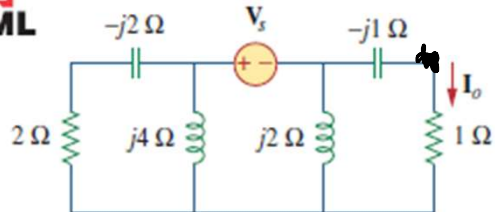
$$Z_2 = (1-j) \parallel j2$$

$$= \frac{(1-j) \times j2}{(1-j) + j2}$$

$$= 2$$



9.54 In the circuit of Fig. 9.61, find V_s if $I_o = 2\angle 0^\circ$ A.



$$V_{(1-j)} = 2\angle 0^\circ \times (1-j) = 2.83\angle -45^\circ$$

$$V = I \times Z$$

$$V_{Z_2} = 2.83\angle -45^\circ = V_1$$

$$V_{z_1} = 2 \times V_{z_2}$$

$$= 2(2.83 \angle -45^\circ)$$

$$= 5.66 \angle -45^\circ$$

$$= V_1$$

$$z_2 = 2z_1$$

↓

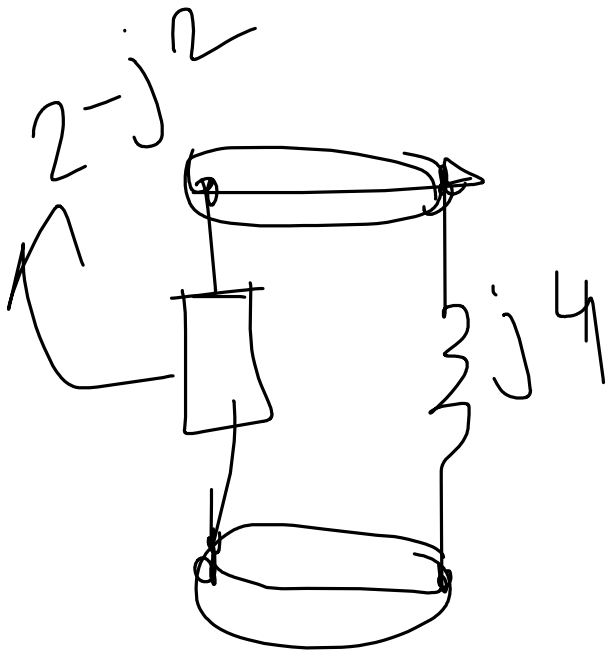
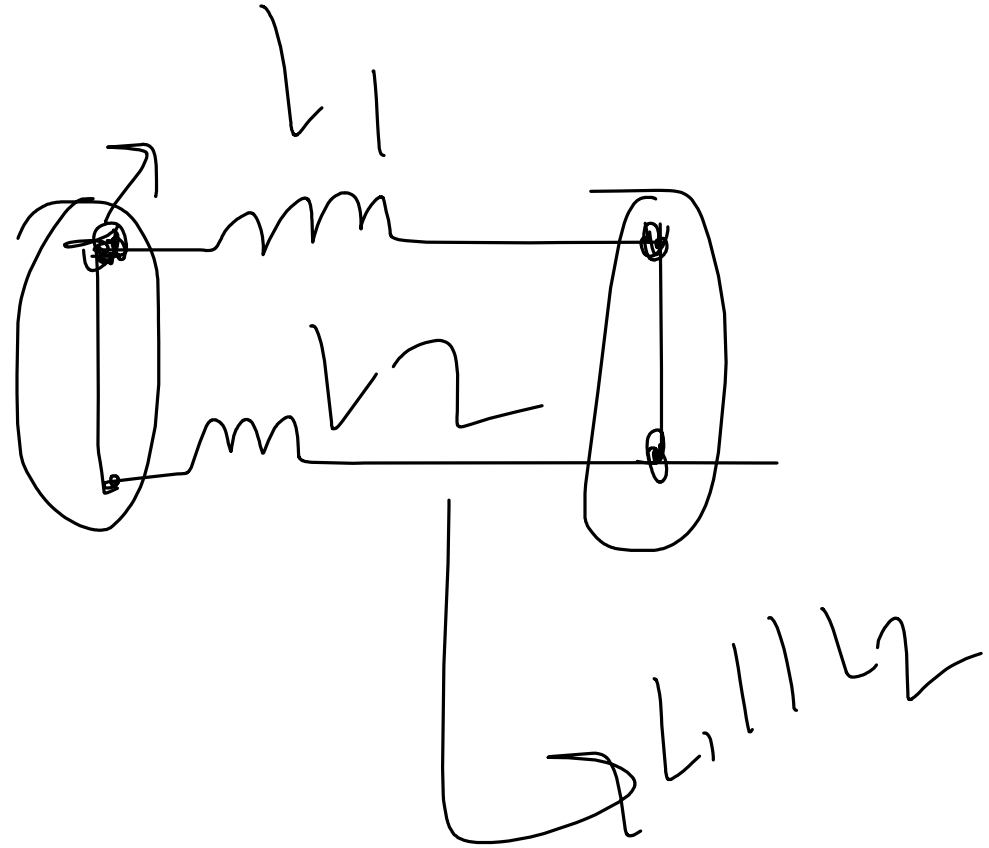
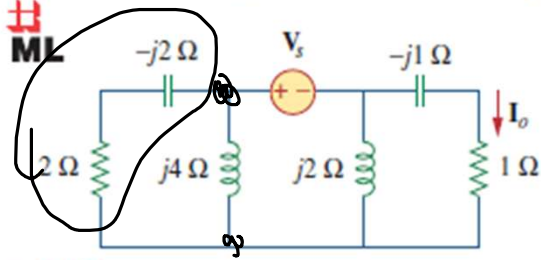
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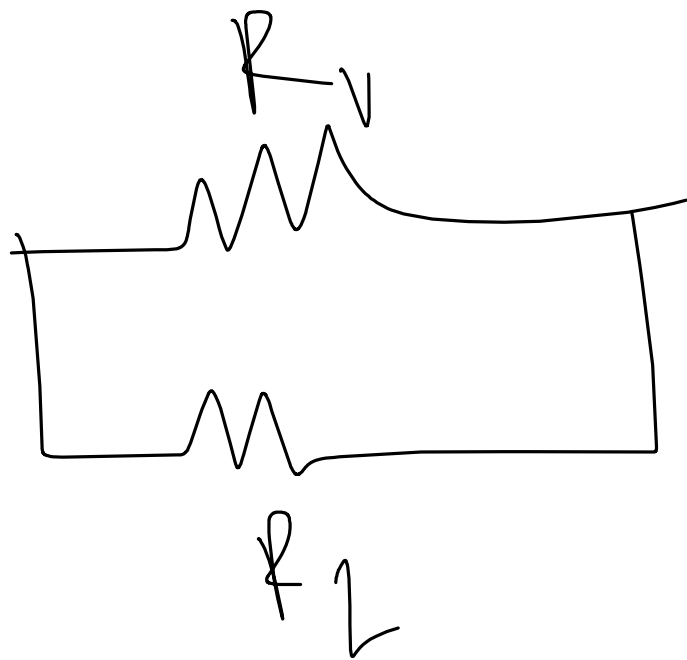
KVL 1

$$V_5 + V_2 + V_1 = 0$$

$$\Rightarrow V_1 = -V_2 - V_1$$
$$= -2 \cdot 43 \angle 45^\circ - 5 \cdot 66 \angle 45^\circ$$
$$= 8.49 \angle 135^\circ$$

9.54 In the circuit of Fig. 9.61, find V_s if $I_o = 2\angle 0^\circ$ A.





$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

↓

$$\left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$